1. Two parts of this problem are independent:

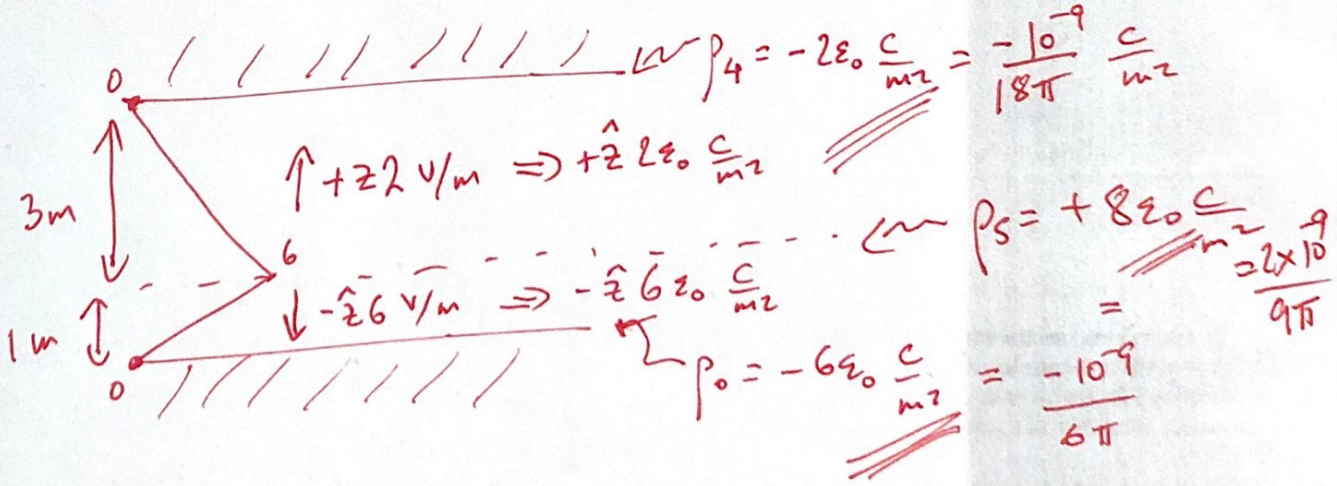
a) (10 pts) Determine the electrostatic potential function $V(x,y,z)$ corresponding to $\mathbf{E} = y\hat{x} + x\hat{y} + \cos(z)\hat{z}$ V/m if $V(0,0,0) = 2$ V.

$$dV = -\bar{E}\cdot d\bar{l} = -(y\,dx + x\,dy + \cos(z)\,dz)$$
$$= -d(xy) - d(\sin(z))$$

$$\hookrightarrow V(x,y,z) = -xy - \sin(z) + C$$
$$V(0,0,0) = C = 2$$

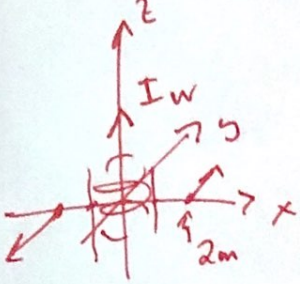
$$\hookrightarrow V(x,y,z) = 2 - xy - \sin(z) \ \text{V}.$$

b) (15 pts) Conducting plates placed on $z = 0$ and $z = 4$ m surfaces (of infinite extent) are separated by vacuum and maintained at potentials of $V(0) = V(4) = 0$ V. Also, the potential of the $z = 1$ m surface that supports a surface charge density $\rho_s$ is $V(1) = 6$ V. Determine $\rho_s$ and the surface charge densities $\rho_0$ and $\rho_4$ at $z = 0$ and $z = 4$ m, respectively, by using the piecewise linear solution of the Laplace's equation for the region.



2. Consider an infinite solenoid having $N = 500$ loops/meter wound around a cylindrical rod of radius of $a = 1$ m and permeability $\mu = 2\mu_o$ carrying a current of $I_s = 2$ mA (milliamp). Also, an infinite wire running along the $z$-axis is carrying a current $I_w = 1$ A in $+z$ direction.

a) (6 pts) Using Ampere's law, determine the **H** vector at position $\mathbf{r} = \hat{x}2$ m outside the solenoid.




$$\bar{B} = \frac{\mu_0 I_w}{2\pi x}\hat{\varphi}\,T \quad (I_w = 1A,\ x = 2m) \Rightarrow \bar{H} = \frac{\hat{\varphi}}{4\pi}\ \frac{A}{m}$$

b) (6 pts) What is vector **H** at position $\mathbf{r} = \hat{x}0.5$ m within the solenoid?

$$\bar{H} = \frac{\hat{\varphi}}{\pi} + \hat{z}$$

$\hat{z}$ due to solenoid, $J_s = 1$ A/m ....

Increased by factor 4 as distance is halved twice

c) (6 pts) Write the expression for vector $\mathbf{B}(x, 0, 0)$ for $|x| < 1$ m (within the solenoid).

$$\bar{B} = \frac{2\mu_0 I_w}{2\pi x}\hat{\varphi} + 2\mu_0\hat{z} = 2\mu_0\left(\frac{1}{2\pi x}\hat{\varphi} + \hat{z}\right)T$$

d) (7 pts) **Essay question:** How does $\mathbf{B}(x, 0, 0)$ vector change across the surface of the solenoid across the location $x = 1$ m? Describe with a few sentences.

$\hat{z}$ comp of $\bar{B}$ will disappear since solenoid fields are internal.

$\hat{\varphi}$ comp of $\bar{B}$ will jump from $\frac{\mu_0}{\pi}$ to $\frac{\mu_0}{2\pi}$ across $x = 1$ m so that $H_y = \frac{1}{2\pi}$ is continuous. This jump is a result of jump in $\mu$!!!

3. Consider a time-varying TEM wave field with $\mathbf{E}(x,t) = \hat{z}\cos(2\pi f t - 2\pi x)$ V/m propagating in free space.

a) (6 pts) Determine the wavelength $\lambda$ and frequency $f$ of the wave field in m and MHz units, respectively.

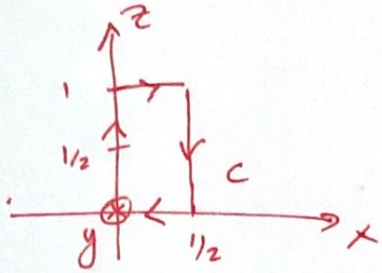
$\beta = 2\pi \Rightarrow \lambda = \frac{2\pi}{\beta} = 1$ m ///.

$\frac{\omega}{\beta} = \frac{2\pi f}{\beta} = c \Rightarrow f = 3\times 10^8$ Hz $= 300$ MHz ///.

b) (4 pts) What is the divergence $\nabla \cdot \mathbf{E}(x,0)$ of the wave field at $t = 0$?

$\nabla \cdot \bar{E} = \frac{\partial}{\partial z}\cos(2\pi f t - 2\pi x) = 0$ //.

c) (8 pts) Given $\mathbf{E}(x,0)$ compute the corresponding emf $\mathcal{E}$ over a closed contour $C$ of a rectangular shape on the $xz-$plane with the opposing corners located at $(x,z) = (0,0)$ and $(x,z) = (\frac{1}{2},1)$ m and traversed in clockwise direction when viewing $C$ looking in the $+y$ direction.



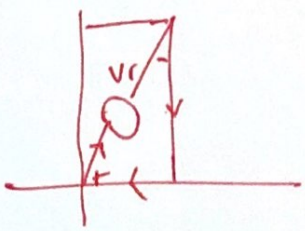

$\mathcal{E} = \oint_C \bar{E}\cdot d\bar{\ell} = \underbrace{E_z(0,0)}_{1}\times 1 + 0 + \underbrace{E_z(\frac{1}{2},0)}_{-1}(-1) + 0$

$= 2$ V //.

d) (4 pts) What would be the voltage reading $V_r$ of a voltmeter at time $t = 0$ connected directly (vertically) between $(x,z) = (\frac{1}{2},0)$ and $(x,z) = (\frac{1}{2},1)$ with the positive probe contacting $(x,z) = (\frac{1}{2},0)$? Explain your reasoning.

$V_r = \int_{(1/2,0)}^{(1/2,1)} \bar{E}\cdot d\bar{\ell} = \int_{(1/2,0)}^{(1/2,1)} (-1\hat{z})\cdot(\hat{z}\,dz) = -1\int_{z=0}^{z=1} dz = -1$ V //.

e) (3 pts) What would be the voltage reading $V_r$ of a voltmeter at time $t = 0$ connected directly (diagonally) between $(x,z) = (0,0)$ and $(x,z) = (\frac{1}{2},1)$ with the positive probe contacting $(x,z) = (0,0)$? **Hint:** you may need to use the result of part (c) in this part.




$V_r + 1 = \frac{\mathcal{E}}{2}$ ← from part (c) this is 2 V

this is by symmetry, half of $\mathcal{E}$ belongs to the pair of two triangles.

$\hookrightarrow V_r = 0$ !!



Alternative derivation:

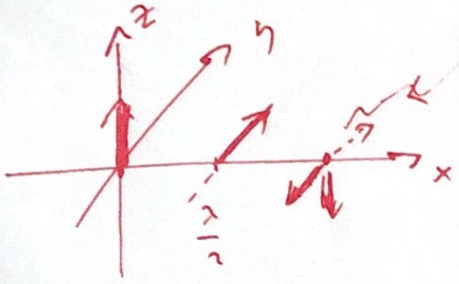
$V_r = \int_{(0,0)}^{(\frac{1}{2},1)} \hat{z}\cos(-2\pi x)\cdot\underbrace{(\hat{x}+2\hat{z})\,dx}_{d\bar{\ell}} = 2\int_0^{1/2} dx\cos(2\pi x) = 0$ //.

4. Two current sheets are fixed in free-space in the $x = 0$ and $x = \lambda/2$ planes having surface current densities $\mathbf{J}_s(t) = \cos(\omega t + \phi)\hat{z}$ and $\mathbf{J}_s(t) = \cos(\omega t)\hat{y}$ A/m, respectively, where $\phi$ is some real number. TEM waves will propagate away from each sheet and superpose with each other to create a wavefield whose polarization depends on $\phi$.

a) (10 pts) If the electric field phasor for $\mathbf{r} = (x, y, z)$ that has been radiated into $x > \frac{\lambda}{2}$ region by these current sheets is expressed as

$$\tilde{\mathbf{E}}(x, y, z) = Pe^{-j\beta x}(\hat{y} + Q\hat{z})$$

determine $P$ and $Q$. **Hint:** $P$ is in V/m while $Q$ is dimensionless and depends on $\phi$.

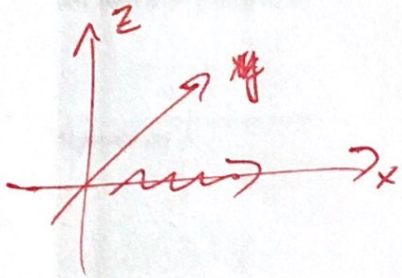



$$\bar{E} = -\hat{z}\frac{\eta_0}{2}e^{j\phi}e^{-j\beta x} - \hat{y}\frac{\eta_0}{2}e^{-j\beta(x-\frac{\lambda}{2})}$$

$$= -\frac{\eta_0}{2}e^{-j\beta x}\left(\hat{z}e^{j\phi} + \hat{y}\underbrace{e^{j\frac{2\pi}{\lambda}\cdot\frac{\lambda}{2}}}_{e^{j\pi}=-1}\right) \Rightarrow P = \frac{\eta_0}{2} = 60\pi \frac{V}{m}$$

$$= +\frac{\eta_0}{2}e^{-j\beta x}(\hat{y} - e^{j\phi}\hat{z}) \qquad Q = -e^{j\phi}$$

b) ( 9 pts) Determine the values of $\phi$ for the superposed TEM waves propagating in the region $x > \lambda/2$ to have:




i. linear polarization,

$\phi = 0, \pi, 2\pi \cdots$

ii. right handed circular polarization,

$\hat{y}$ leads $\hat{z}$ $\Rightarrow$ $\phi = \frac{\pi}{2}$

iii. left handed circular polarization.

$\hat{z}$ leads $\hat{y}$ $\Rightarrow$ $\phi = -\frac{\pi}{2}$

c) (6 pts) Determine the time-averaged Poynting vector in the region $x > \lambda/2$ with $\phi$ determined for all 3 cases in part (b):

$$\bar{E} = P e^{-j\beta x}(\hat{y} + Q\hat{z})$$

$$\bar{H} = \frac{P e^{-j\beta x}}{\eta_0}(\hat{z} - Q\hat{y})$$

i. linear polarization,

$\langle S \rangle = 30\pi\,\hat{x}$ W/m$^2$

ii. right handed circular polarization,

same

iii. left handed circular polarization.

same

$$\langle S \rangle = \frac{1}{2}\mathrm{Re}\{\bar{E} \times \bar{H}^*\}$$

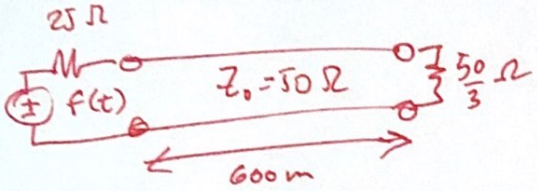
$$= \frac{P^2}{2\eta_0}(\hat{x} + \hat{x}QQ^*)$$

$$= \frac{P^2}{\eta_0}\hat{x}$$

$$= \frac{60\pi \times 60\pi}{120\pi}\hat{x} = 30\pi\,\hat{x}$$

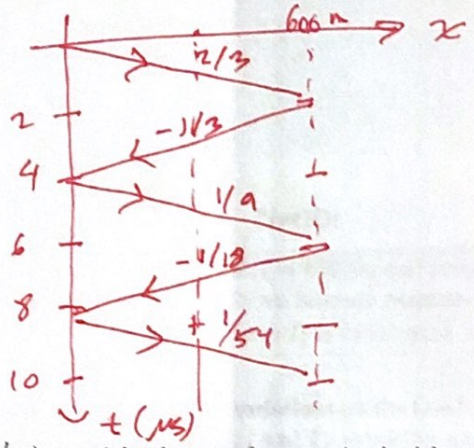
5. Consider a transmission line with characteristic impedance $Z_o = 50\,\Omega$, length $l = 600$ m, and propagation velocity $v = \frac{1}{\sqrt{LC}} = c = 3 \times 10^8$ m/s. A voltage source $f(t)$ with an internal resistance $R_g = 25\,\Omega$ is connected to one end of the T.L. (at $z = 0$) and the other end ($z = l$) is terminated by a load resistance $R_L = 50/3\,\Omega$.

a) (10 pts) Construct a bounce diagram to determine the voltage, $V(z,t)$, variations on the line for $0 < z < l$ and $0 < t < 10\,\mu$s for $f(t) = \delta(t)$. Write the values of $\tau_g$, $\Gamma_L$, and $\Gamma_g$ explicitly and make sure to label both axis of the bounce diagram properly in terms of m and $\mu$s units and indicate the amplitudes of traveling impulses next to the impulse trajectory lines.




$\Gamma_g = \frac{25-50}{25+50} = \frac{-25}{75} = -\frac{1}{3}$ $\quad \Gamma_L = \frac{\frac{50}{3}-50}{\frac{50}{3}+50}$

$\Gamma_g = \frac{50}{25+50} = \frac{50}{75} = \frac{2}{3}$ $\quad = \frac{1-3}{1+3} = -\frac{1}{2}$

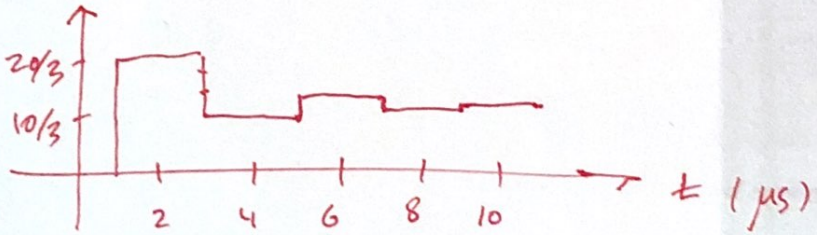



b) (4 pts) Write your expressions for $V(\frac{l}{2},t)$ and $I(\frac{l}{2},t)$ as weighted sums of appropriately delayed impulses $\delta(t)$ for $0 < t < 10\,\mu$s .

$$V(300,t) = \frac{2}{3}\delta(t-1) + \left(\frac{-1}{3}\right)\delta(t-3) + \left(\frac{1}{9}\right)\delta(t-5) + \left(\frac{-1}{18}\right)\delta(t-7) + \frac{1}{54}\delta(t-9)$$

$$I(300,t) = \frac{1}{50}\left[\frac{2}{3}\delta(t-1) + \frac{1}{3}\delta(t-3) + \frac{1}{9}\delta(t-5) + \frac{1}{18}\delta(t-7) + \frac{1}{54}\delta(t-9)\right]$$

c) (6 pts) Plot $V(\frac{l}{2},t)$ as a function of $t$ for $0 < t < 10\,\mu$s if $f(t) = 10u(t)$ V.




d) Determine the steady state voltage $V_L$ and current $I_L$ on the line for the excitation given in part (c) as time $t \to \infty$.

As $t \to \infty$ DC steady state and using voltage division

$$V_L = 10\,\frac{50/3}{25+50/3} = 10\,\frac{50}{75+50} = 10\,\frac{50}{125} = \frac{500}{125} = 4\text{ V}.$$

$$I_L = \frac{4\text{V}}{50/3\,\Omega} = \frac{12}{50} = \frac{24}{100} = 0.24\text{ A}.$$